

# Flask Web Framework: Detailed Study Notes (Roman Urdu)

## 1 Flask Kya Hai

Flask ek lightweight micro web framework hai jo Python me web applications banane ke liye use hota hai. Micro ka matlab hai ke isme sirf basic features hote hain aur baqi cheezein developer khud add karta hai. Yeh flexibility deta hai aur control bhi zyada deta hai.

Flask zyada tar use hota hai:

- Web development
- APIs banane ke liye
- Data science dashboards
- Chhote aur medium projects

## 2 Installation

Flask install karne ke liye yeh command use hoti hai:

```
pip install flask
```

## 3 Basic Flask Application

```
from flask import Flask

app = Flask(__name__)

@app.route("/")
def home():
    return "Hello World"

app.run(debug=True)
```

**Explanation:**

- `Flask(__name__)` Flask app create karta hai
- `@app.route("/")` URL define karta hai
- `home()` function response return karta hai
- `debug=True` errors show karta hai

## 4 Routing

Routing ka matlab hai URL ko function ke sath connect karna.

```
@app.route("/about")
def about():
    return "About Page"
```

### 4.1 Dynamic Routing

```
@app.route("/user/<name>")
def user(name):
    return f"Hello {name}"
```

Isme user ka naam URL se liya jata hai.

## 5 Request Handling

Flask GET aur POST requests handle karta hai.

```
from flask import request

@app.route("/data", methods=["GET", "POST"])
def data():
    if request.method == "POST":
        return "POST Request"
    return "GET Request"
```

## 6 Templates (HTML)

Flask Jinja2 template engine use karta hai.

HTML file (templates/index.html):

```
<h1>Hello {{ name }}</h1>
```

Python code:

```
from flask import render_template

@app.route("/")
def home():
    return render_template("index.html", name="Ali")
```

**Explanation:**

- render\_template HTML ko load karta hai
- name dynamic data show karta hai

## 7 Forms Handling

HTML Form:

```
<form method="POST">
    <input type="text" name="username">
    <input type="submit">
</form>
```

Flask code:

```
@app.route("/form", methods=["GET", "POST"])
def form():
    if request.method == "POST":
        name = request.form["username"]
        return f"Hello {name}"
    return render_template("form.html")
```

## 8 Project Structure

project/

```
app.py
templates/
    index.html
    form.html
static/
```

## 9 Session Management

Session user ka data temporarily store karta hai.

```
from flask import session

app.secret_key = "secret"

@app.route("/login")
def login():
    session["user"] = "Ali"
    return "Logged in"
```

## 10 Error Handling

```
@app.errorhandler(404)
def not_found(e):
    return "Page Not Found", 404
```

## 11 Flask Extensions

Flask ko powerful banane ke liye extensions use hote hain:

- Flask-SQLAlchemy (database)
- Flask-Login (authentication)
- Flask-WTF (forms)

## 12 Basic Project: Student Registration App

### 12.1 Project Overview

Is project me user apna data form me enter karta hai aur submit karne ke baad next page par show hota hai.

### 12.2 Project Structure

flask\_project/

```
app.py
templates/
    home.html
    result.html
```

### 12.3 HTML Form (home.html)

```
<!DOCTYPE html>
<html>
<body>
    <h2>Student Form</h2>
    <form method="POST">
        Name: <input type="text" name="name"><br><br>
        Age: <input type="number" name="age"><br><br>
        <input type="submit">
    </form>
</body>
</html>
```

### 12.4 Result Page (result.html)

```
<!DOCTYPE html>
<html>
<body>
    <h2>Student Data</h2>
    <p>Name: {{ name }}</p>
    <p>Age: {{ age }}</p>
</body>
</html>
```

### 12.5 Flask Code (app.py)

```
from flask import Flask, render_template, request

app = Flask(__name__)

@app.route("/", methods=["GET", "POST"])
def home():
    if request.method == "POST":
        name = request.form["name"]
        age = request.form["age"]
        return render_template("result.html", name=name, age=age)
    return render_template("home.html")
```

```
app.run(debug=True)
```

## 12.6 Working Explanation

- User homepage open karta hai
- Form fill karta hai
- Submit karta hai
- Flask data receive karta hai
- Data next page par show hota hai

## 13 Conclusion

Flask ek simple aur flexible framework hai jo beginners ke liye easy hai aur professionals ke liye powerful hai. Yeh chhote projects aur APIs ke liye best hai.